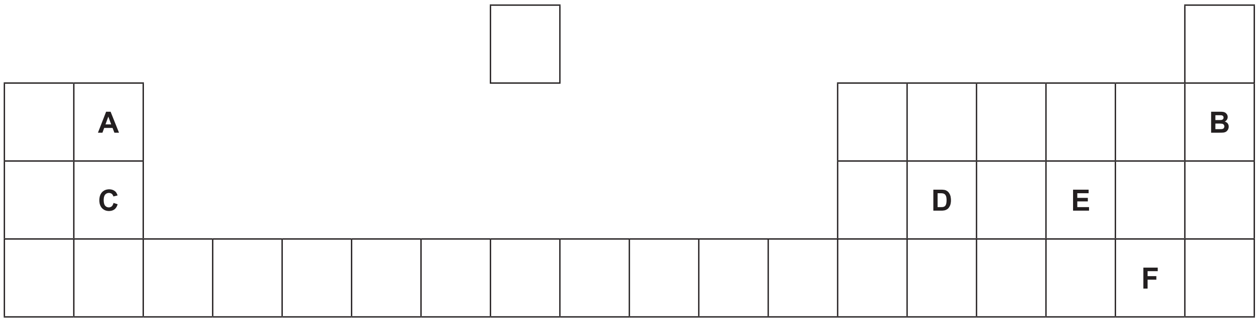
8. *(a)* The following diagram shows an outline of part of the Periodic Table.

The letters shown are NOT the chemical symbols of the elements.

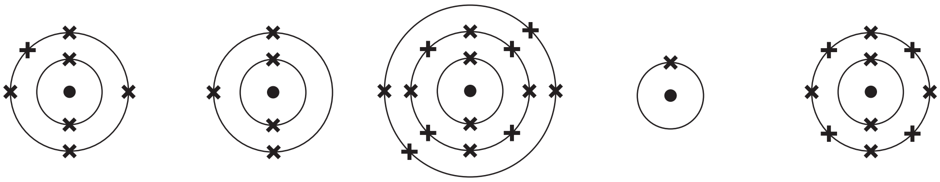


Choose **letters** from the diagram to complete the table below. [5]

	Letter(s)
Two elements in the same group	.... and ....
The element which has 12 protons in its nucleus	....
The element with a full outer shell of electrons	....
The element in Group 2 and Period 2	....
The element with the electronic structure 2,8,4	....

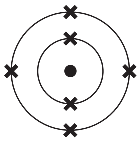
Examiner only

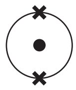
(b) Diagrams **1-5** show the electronic structure of five elements in the Periodic Table.



1 **2** **3** **4** **5**

Give the **number** of the diagram which shows the electronic structure of the element which lies

(i) directly **below**  in the Periodic Table, ... [1]

(ii) to the **left** of  in the Periodic Table. .... [1]

(c) Nitrogen has two stable isotopes – nitrogen-14 and nitrogen-15.

Describe how these isotopes are similar to one another and how they are different. [2]

....

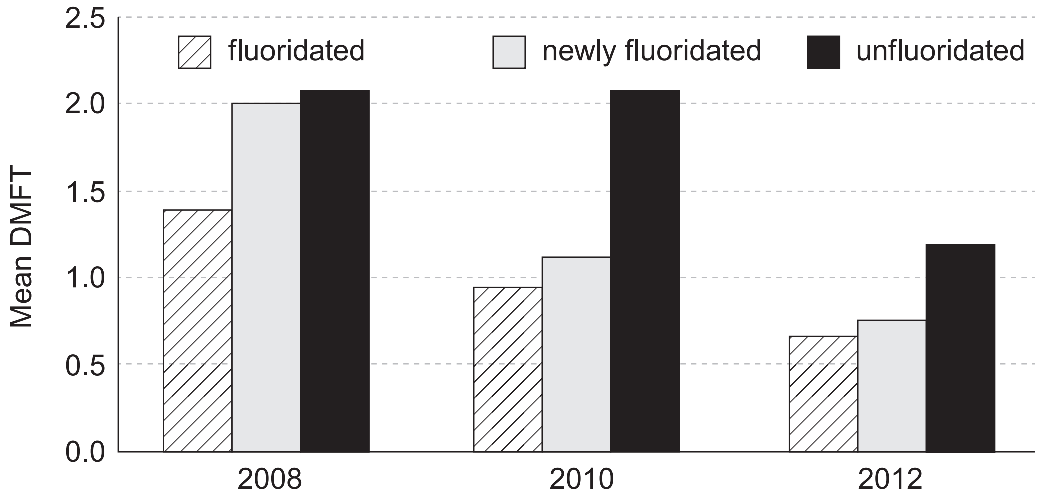
....

....

9

Examiner only

8. The following graph shows the mean numbers of decayed, missing and filled teeth (DMFT) in 12 year-olds in three areas of Australia in 2008, 2010 and 2012. One area has been fluoridated for 20 years, one is newly fluoridated and the other is unfluoridated.



Describe what the graph tells us. Use this information and your knowledge of fluoridation to explain why some people support fluoridation of water supplies but others oppose it. [6 QER]

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Examiner only

8. *(a)* A student was investigating how the concentration of sodium thiosulfate solution affects its reaction with hydrochloric acid.

(i) The reaction taking place can be represented by the following equation, which is **not** balanced.

Write the number **2** in **one of the boxes** on the right hand side in order to balance the equation. [1]

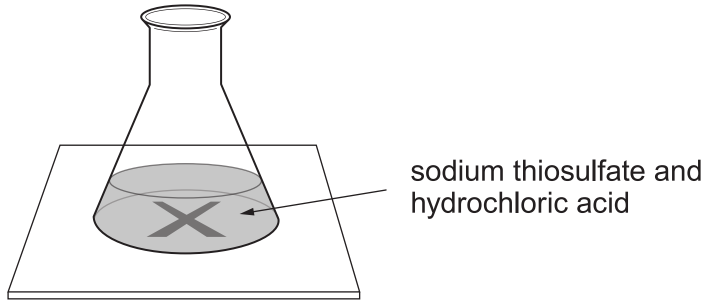
$$2HCl(aq) + Na_2S_2O_3(aq) \longrightarrow \boxed{}\,SO_2(g) + \boxed{}\,S(s) + \boxed{}\,NaCl(aq) + \boxed{}\,H_2O(l)$$

(ii) During the reaction, the solution becomes cloudy due to the formation of a precipitate. State the meaning of the term *precipitate*. [1]

....

....

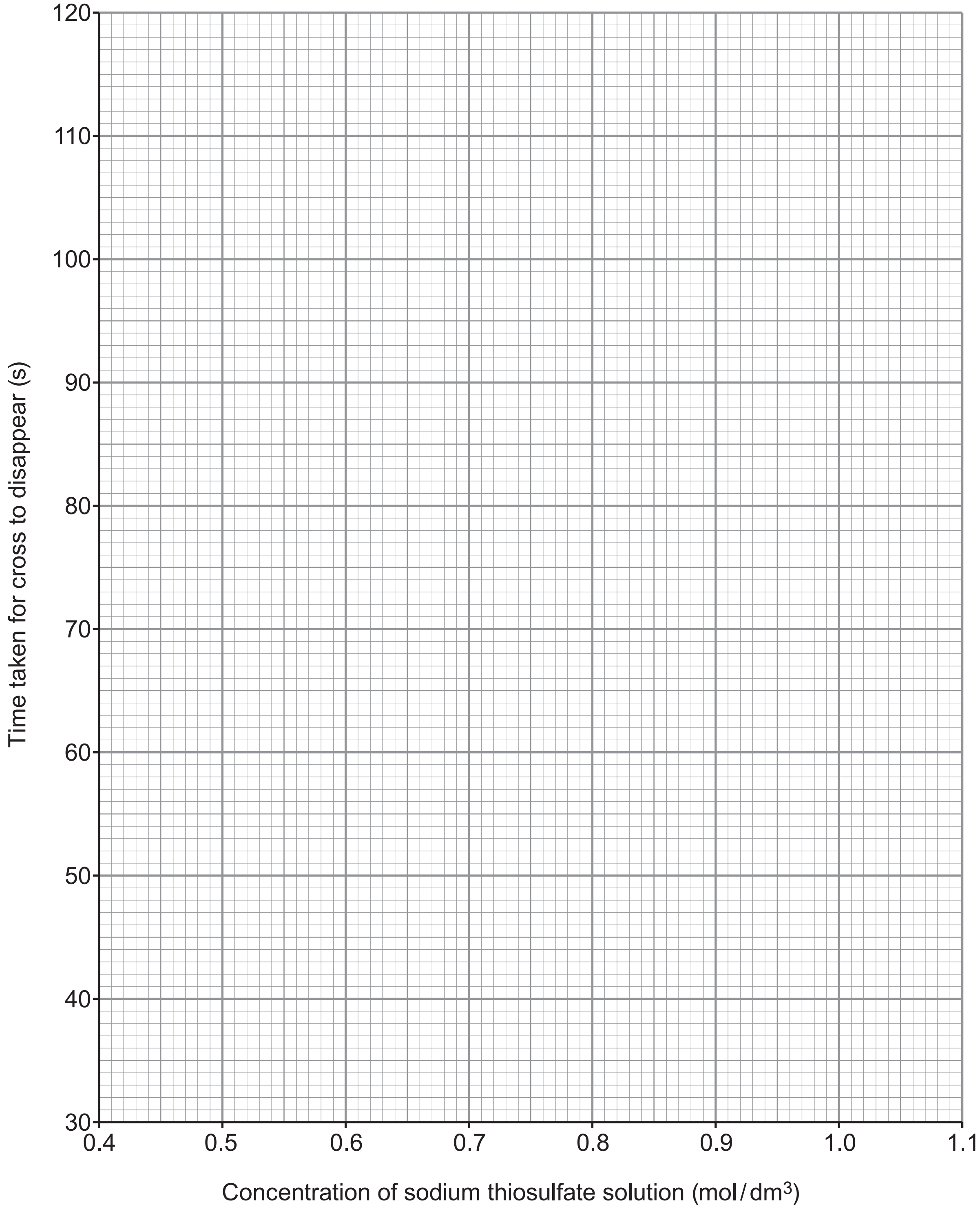
(iii) The time taken for the cross to disappear was measured as shown below.



The results are shown in the table.

Concentration of sodium thiosulfate solution (mol/dm^3)	Time taken for cross to disappear (s)
1.0	42
0.9	46
0.8	53
0.7	66
0.6	87
0.5	110

Plot the results from the table on the grid. Draw a suitable line. [3]



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(iv) State how concentration affects the **time taken** for the cross to disappear. [1]

....

....

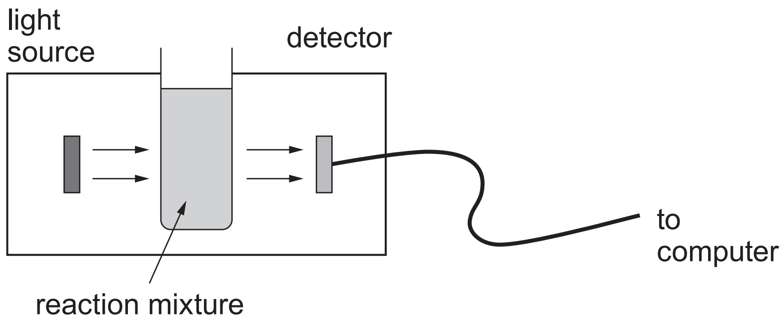
(v) What conclusion can be drawn about the effect of concentration on the **rate** of this reaction? [1]

....

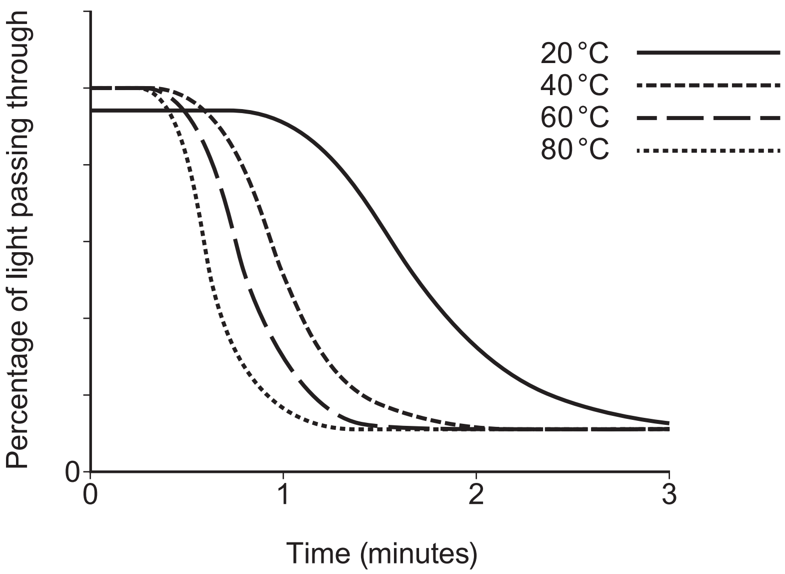
....

Examiner only

(b) A second student used a light sensor to investigate the effect of temperature on the rate of this reaction.



He obtained the following results.



(i) Give **one** conclusion that can be drawn from these results. State how this is shown by the graph. [2]

....

....

....

(ii) Slightly less light passed through the tube at the start of the experiment at 20 °C than in the others. Suggest a possible practical reason for this. [1]

....

....

10

Examiner only

8. Burning fossil fuels leads to global warming.

Explain how global warming arises and describe the problems caused. [6 QER]

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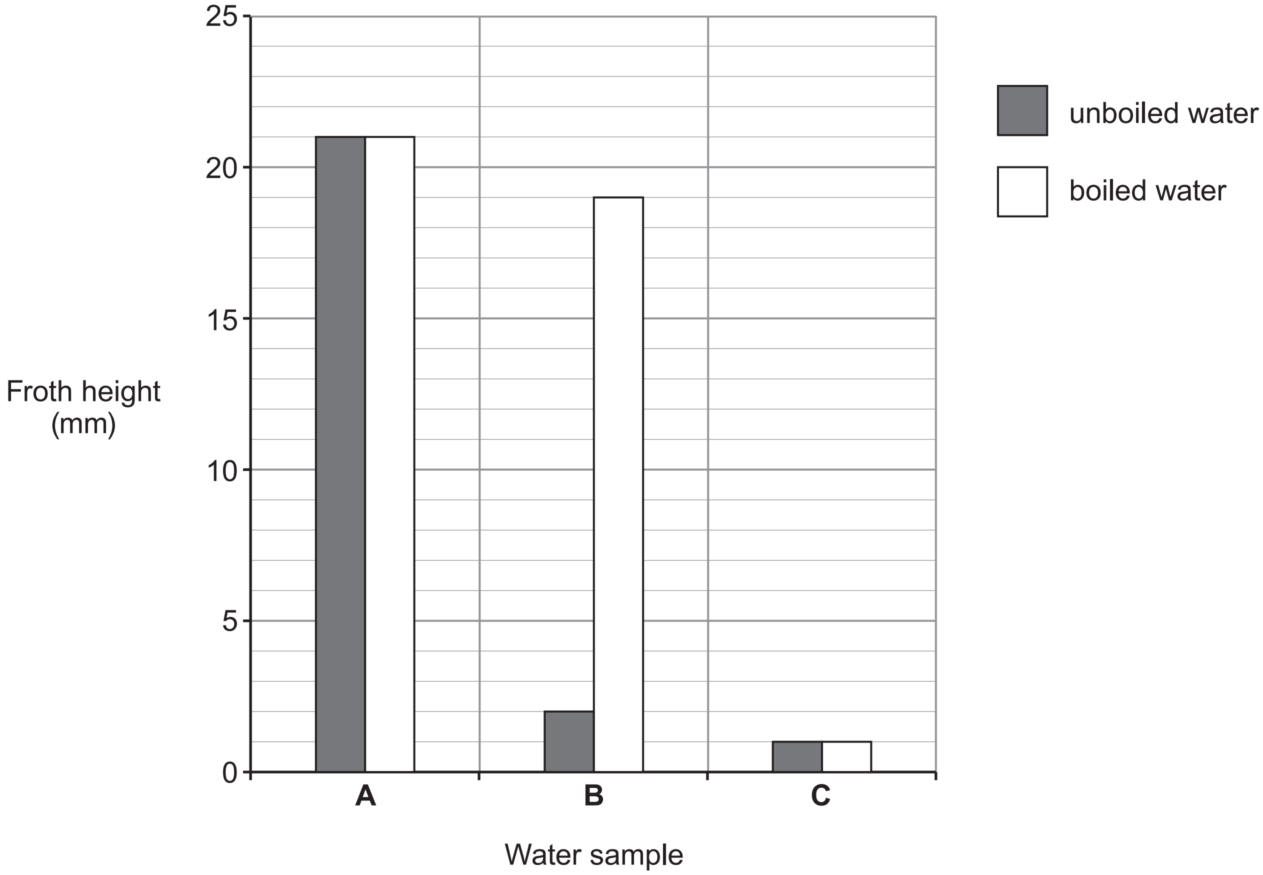
Examiner only

8. (a) Three samples of water, **A**, **B** and **C**, from different parts of the UK were tested in a laboratory.

$1\,cm^3$ of soap solution was added to $25\,cm^3$ of the three different water samples. Each sample was shaken for 1 minute. The height of the froth was measured.

The experiment was repeated using new samples of water, **A**, **B** and **C**, that had been boiled.



Give the **letter** of the water sample which is [2]

temporary hard water ....

permanent hard water ....

soft water ....

Examiner only

(b) There are advantages and disadvantages of living in a hard water area.

Give **two** disadvantages of living in a hard water area. [2]

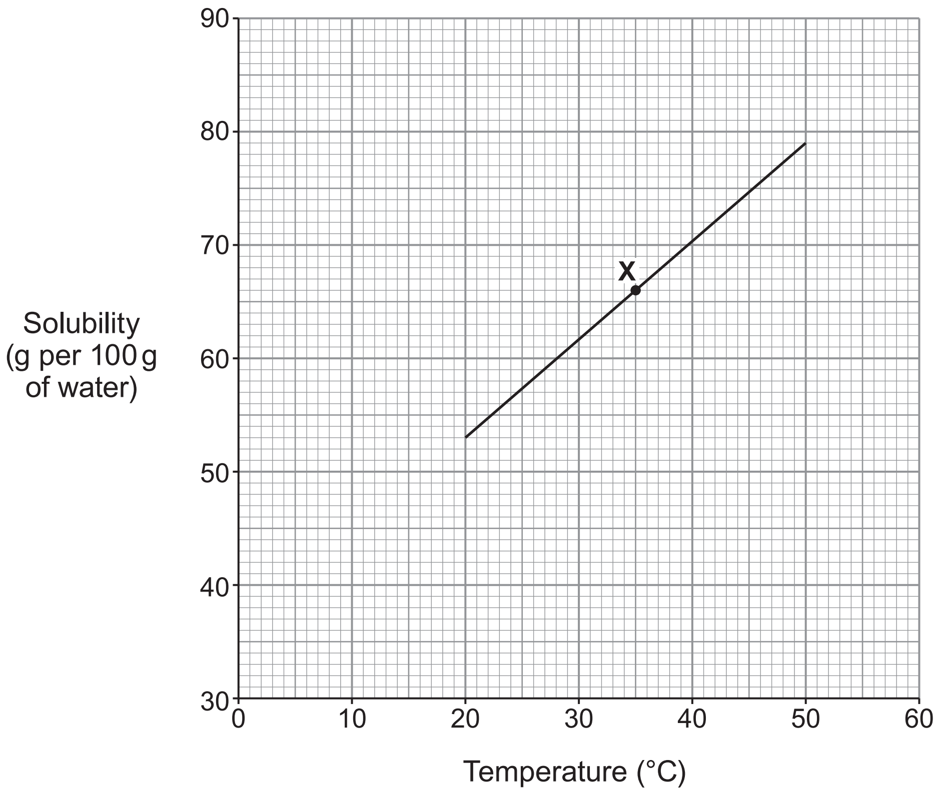
1 ....

....

2 ....

....

(c) The graph below shows the solubility of lead nitrate in water at different temperatures.



(i) State what point **X** on the graph tells you about lead nitrate. [1]

....

....

Examiner only

(ii) The solubility of lead nitrate at 20 °C is 53 g per 100 g of water.

Use the graph to find its solubility at 50 °C and hence calculate the mass of lead nitrate crystals that form when a saturated solution containing 100 g of water cools from 50 °C to 20 °C. [2]

Mass = ... g

(iii) Use the graph to find the solubility of lead nitrate at 5 °C. [1]

.... g per 100 g of water

8